Fast Failure Detection in LLM-generated patches

Background

What is an LLM agent → html page

How it works for a software engineering task:

- LLM agent gets the natural language description of the bug you want to solve
- It acts in a loop until it believes the bug is solved

Do you know traditional ML methods ?

- e.g. Logistic Regression, Random Forest, SVMs, ...
- everything is ready in sklearn

(Possible) Initial Steps

- SWE-Smith → https://huggingface.co/datasets/SWE-bench/SWE-smith-trajectories
  - Understand how the dataset is done. Print statistics, look for null/problematic values. Assess if we can safely use this as a dataset for our classifier.
  - Think about key features that we can extract to build a traditional ML classifier
  - Train a simple ML model as a starting point.
  - **Input features**: extracted from ***Trajectory or Messages/Patch***
  - **Target label to predict**: ***Resolved***
    - **Without** running any unit test

Trajectory / Messages	**Patch**	**Resolved**
Description of the bug to solve + List of agent steps to generate the patch	*Agent-generated patch*	*False/True*

SWE-Smith row important columns

Next steps

- Understand SWE-smith (statistics on the dataset)
- Extract some features to build a dataset
- Train a simple Logistic Regression Model and show results on (train-validation-test)

Extracted_Feat_1	**Extraxted_Feat_i**	**Extraxted_Feat_N**	**Resolved (Target Label)**
			False/True